ALGEBRA QUALIFYING EXAMINATION

Fall 2023

PLEASE WRITE YOUR UO ID NUMBER ON YOUR SOLUTIONS but not your name.

There are three parts to this exam:

- Part I is asking for statements of some definitions/theorems. Three questions, 6 points each, give concise but accurate answers.
- Part II is True/False. Five questions, 8 points each, give a sketch proof or counterexample for full credit.
- Part III is longer problems. Three questions, 14 points each, give concise but clear proofs stating any significant results you are using.

ATTEMPT ALL QUESTIONS

Part I. Briefly state each of the following (6 points each):

1. The definition of an adjoint pair of functors.

Solution. Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. The functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ are adjoint (F is left adjoint of G and G is right adjoint of F) if there exists a natural (in variables $X \in \mathcal{C}, Y \in \mathcal{D}$) isomorphism

$$\mathrm{Hom}_{\mathcal{D}}(FX, Y) \simeq \mathrm{Hom}_{\mathcal{C}}(X, GY).$$

(The definition of naturality could also be spelled out, but we don't insist on this.)

2. The Lying Over Theorem (in commutative algebra).

Solution. Let $R \subset S$ be an integral extension of commutative rings. Then for any prime ideal $p \subset R$ there exists a prime ideal $P \subset S$ lying over p (this means that $P \cap R = p$). More generally for any ideal $I \subset S$ with $I \cap R \subset p$ there exists a prime ideal of S containing I and lying over p.

3. The Jacobson Density Theorem.

Solution. Let R be a ring and let V be an irreducible R-module. Let $D = \mathrm{End}_R(V)$ and let $v_1, \ldots, v_n$ be elements of V linearly independent over D (this makes sense since D is a division algebra by Schur's lemma). Then for arbitrary elements $w_1, \ldots, w_n$ there exists $r \in R$ such that $rv_i = w_i$ for all $i = 1, \ldots, n$.

Part II. True or false? If true provide a brief explanation, if false provide a counterexample (8 points each):

1. There is an isomorphism $D_6 \cong C_2 \times D_3$. Here, C_n is the cyclic group of order n and D_n is the dihedral group of order $2n$.

Solution. TRUE. We have that $D_6 = \langle a, b \mid a^6 = b^2 = 1, bab = a^{-1} \rangle$. Let $A := a^2, B := b$ and $x := a^3$. Then D_6 is also generated by A, B and x. These satisfy the relations $A^3 = B^2 = 1, BAB = A^{-1}$, $xA = Ax, xB = Bx$ and $x^2 = 1$.
We have that $C_2 \times D_3 = \langle x, A, B \mid A^3 = B^2 = 1, BAB = A^{-1}, xA = Ax, xB = Bx, x^2 = 1\rangle$. There is a surjective homomorphism $\theta : C_2 \times D_3 \twoheadrightarrow D_6$ taking x, A and B to the elements of D_6 defined in the previous paragraph. This follows from the relations checked in that paragraph.
Finally we observe that $|C_2 \times D_3| = 12 = |D_6$ so θ is an isomorphism.

Comment: Geometrically, one could just draw a regular hexagon and an equilateral triangle on its even vertices. Then x is "reflection in the origin".

2. Any field extension E/F of degree 2 is Galois.

Solution. FALSE. There exist non-separable extensions of degree 2. For example take $F = \mathbb{F}_2(t)$ (purely transcendental extension of the field with 2 elements) and $E = F[x]/(x^2 - t)$.

3. The $\mathbb{C}$-algebras $\mathbb{C}[x]/(x^3 - x)$ and $\mathbb{C}[x]/(x^3 + x + 1)$ are isomorphic.

Solution. TRUE. Both polynomials have 3 distinct roots. One way to see this in the second case is to observe that $GCD(f(x), f'(x)) = 1$. Now for any polynomial $p(x) \in \mathbb{C}[x]$ of degree 3 with 3 distinct roots $\alpha_1, \alpha_2, \alpha_3$ we have $\mathbb{C}[x]/(p(x)) \simeq \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}$. This follows by the Chinese Remainder Theorem. Here is a direct proof: we have a homomorphism $\mathbb{C}[x] \to \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}$ sending x to $(\alpha_1, \alpha_2, \alpha_3)$; it is clear that its kernel consists of polynomials annihilating $\alpha_1, \alpha_2, \alpha_3$, hence coincides with $(p(x))$. Thus we have an injective homomorphism $\mathbb{C}[x]/(p(x)) \to \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}$. Since both algebras are of dimension 3, it must be surjective and we are done.

4. There exists a non-zero natural transformation $\mathrm{Id} \to \bigwedge^2$, where Id is the identity endofunctor of the category $\mathcal{V}ec_{\mathbb{R}}$ of real vector spaces, and $\bigwedge^2 : \mathcal{V}ec_{\mathbb{R}} \to \mathcal{V}ec_{\mathbb{R}}$ is the functor sending a vector space to its exterior square.

Solution. FALSE. Let $\alpha : \mathrm{Id} \to \bigwedge^2$ be a natural transformation. Then for a one-dimensional space W, we have $\alpha_W = 0$ since $\bigwedge^2 W = 0$. Now for arbitrary V and a vector $v \in V$, let $f : W \to V$ be the linear map sending the basis vector w of W to v. Then $\alpha_V(v) = \alpha_V(f(w)) = \left(\bigwedge^2 f\right)(\alpha_W(w)) = 0$ by the definition of the natural transformation. Thus $\alpha_V = 0$ for any V, hence $\alpha = 0$.

Comment: Combining symmetric, exterior, and tensor powers one can construct many endofunctors of $\mathcal{V}ec_{\mathbb{R}}$, e.g. $V \mapsto V^{\otimes 5} \otimes \bigwedge^3 V \otimes S^2(\bigwedge^4 V)$. One can also compute natural transformations between such functors using the same approach as above (try to do it, say for $\bigwedge^2 \to \mathrm{Id}$). This is the starting point of the theory of *polynomial functors*.

5. Any finite ring R (i.e., a ring of cardinality $|R| < \infty$) with no non-zero nilpotent elements is commutative.

Solution. TRUE. Let R be a finite ring without nilpotents. Then R is artinian (since it is finite), so its Jacobson radical is nilpotent, hence zero (since R has no nilpotent elements). Thus R is semisimple, so it factors as a product of matrix rings over division algebras by Artin-Wedderburn theorem. However any matrix ring of size > 1 contains nilpotents, e.g. $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, so R is isomorphic to a direct product of finite division algebras. Finally by Wedderburn theorem any finite division algebra is a field and we are done.

Part III. Give complete solutions to the following (14 points each):

1. Let G be a finite group and $A \leq G$ be an Abelian subgroup.

   (a) Prove that all irreducible $\mathbb{C}A$-modules are one-dimensional.

   (b) Prove that the dimension of an irreducible $\mathbb{C}G$-module is at most $[G : A]$.

Solution. (a) Here are three different ways to explain this.

- As $\mathbb{C}A$ is a finitely generated commutative $\mathbb{C}$-algebra, it is a quotient of the polynomial algebra $\mathbb{C}[x_1, \dots, x_n]$ for some $n \geq 1$. Thus, it suffices to prove that all irreducible $\mathbb{C}[x_1, \dots, x_n]$-modules are one-dimensional. Up to isomorphism, irreducible $\mathbb{C}[x_1, \dots, x_n]$-modules are the quotients $\mathbb{C}[x_1, \dots, x_n]/I$ by maximal ideals I. By the Nullstellensatz, the maximal ideals are $(x_1 - c_1, \dots, x_n - c_n)$ for points $(c_1, \dots, c_n) \in \mathbb{C}^n$. Since $(x_1 - c_1, \dots, x_n - c_n)$ is the kernel of the homomorphism $\mathbb{C}[x_1, \dots, x_n] \twoheadrightarrow \mathbb{C}, x_i \mapsto c_i$, we have that $\mathbb{C}[x_1, \dots, x_n]/(x_1 - c_1, \dots, x_n - c_n) \cong \mathbb{C}$, which is one-dimensional.

- By Maschke's theorem, $\mathbb{C}A$ is semisimple. Hence, by Artin-Wedderburn, we have that $\mathbb{C}A \cong M_{d_1}(\mathbb{C}) \oplus \cdots \oplus M_{d_n}(\mathbb{C})$ where n is the number of isomorphism classes of irreducible $\mathbb{C}A$-modules and $d_1, \dots, d_n$ are the dimensions of these modules. Since $\mathbb{C}A$ is commutative we must have that $d_i = 1$ for each i (and $n = |A|$).

- Let $\chi_1, \dots, \chi_n$ be the irreducible characters of A. We have that

$$\chi_1(1)^2 + \cdots + \chi_n(1)^2 = |A|.$$

  The number n of rows in the character table is equal to the number of columns, that is, the number of conjugacy classes in A. Since A is commutative, all its conjugacy classes are of size 1, so $n = |A|$. Thus, $\chi_1(1)^2 + \cdots + \chi_n(1)^2 = n$, so we must have that $\chi_i(1) = 1$ for each i.

(b) Let V be an irreducible $\mathbb{C}G$-module. Since irreducible $\mathbb{C}A$-modules are one-dimensional by (a), V contains a one-dimensional subspace, say $\mathbb{C}v$ invariant under A. The $\mathbb{C}$-span of vectors $\{gv\}_{g\in G}$ is clearly a G-invariant subspace of V, hence coincides with V since V is irreducible. Observe that for any $a \in A$ the vectors gv and gav are proportional (since av is proportional to v). Thus the set $\{gv\}_{g\in G}$ contains at most $[G : A]$ non-proportional vectors (since every right A-coset gives the same vector up to a scalar). It follows that dimension of the span of $\{gv\}_{g\in G}$ is $\leq [G : A]$.

Comment: Here is a more abstract version of the proof of (b). There exists a one-dimensional $\mathbb{C}A$-module W and nonzero homomorphism (of $\mathbb{C}A$-modules) $W \to \mathrm{Res}^G_A V$. Thus by Frobenius reciprocity we have a nonzero homomorphism (of $\mathbb{C}G$-modules) $\mathrm{Ind}^G_A W \to V$. This homomorphism must be surjective since its image is a non-zero $\mathbb{C}G$-submodule of V. Thus $\dim(V) \leq \dim(\mathrm{Ind}^G_A W) = [G : A]\dim(W) = [G : A]$.

2. This is a question about matrices over principal ideal domains.

(a) Compute the invariant factors $\delta_1(x)|\delta_2(x)|\delta_3(x)$ of the matrix

$$\begin{pmatrix} x & -1 & 0 \\ -1 & x & -1 \\ 0 & -1 & x \end{pmatrix} \in \mathrm{Mat}_3(\mathbb{Q}[x]).$$

Hence, write down the minimal polynomial of the matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

over the field $\mathbb{Q}$.

(b) Prove that any matrix $A \in \mathrm{Mat}_n(\mathbb{Z})$ of determinant 6 can be factored as $A = BC$ for $B, C \in \mathrm{Mat}_n(\mathbb{Z})$ with $\det(B) = 2$, $\det(C) = 3$.

Solution. (a) We just have to do some elementary row and column operations:

$$\begin{pmatrix} x & -1 & 0 \\ -1 & x & -1 \\ 0 & -1 & x \end{pmatrix} \sim \begin{pmatrix} 0 & x^2-1 & -x \\ -1 & x & -1 \\ 0 & -1 & x \end{pmatrix} \sim \begin{pmatrix} 0 & x^2-1 & -x \\ -1 & 0 & 0 \\ 0 & -1 & x \end{pmatrix}$$
$$\sim \begin{pmatrix} 0 & x^2-1 & x^3-2x \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & 0 & x^3-2x \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & x^3-2x \end{pmatrix}$$

It follows that the invariant factors are $1, 1, x^3 - 2x$. The minimal polynomial of the second matrix is $\delta_3(x) = x^3 - 2x$.

(b) There exist invertible matrices $P, Q \in GL_n(\mathbb{Z})$ such that $PAQ = D$ where $D = \mathrm{diag}(d_1, \ldots, d_n)$ for non-negative integers $d_1 | \cdots | d_n$ ("Smith normal form"). We are given that A has determinant 6. Since P and Q have determinant ± 1, we deduce that $\prod_{i=1}^n d_i = 6$. Hence, $d_1, \ldots, d_{n-1} = 1$ and $d_n = 6$. So $D = D_1 D_2$ where $D_1 := \mathrm{diag}(1, \ldots, 1, 2)$ and $D_2 := \mathrm{diag}(1, \ldots, 1, 3))$. If $\det P = \det Q = 1$ then we obtain the desired factorization $A = BC$ by setting $A := P^{-1} D_1$ and $B := D_2 Q^{-1}$. Otherwise, $\det P = \det Q = -1$, and such a factorization is given by $A = BC$ with $B := P^{-1} D_1', C := D_2' Q^{-1}$ for $D_1' := \mathrm{diag}(1, \ldots, 1, -2)$ and $D_2' := \mathrm{diag}(1, \ldots, 1, -3)$.

3. All algebraic sets in this question are defined over the field $\mathbb{C}$. Suppose that X and Y are *isomorphic* algebraic sets, and that X has precisely two irreducible components X_1 and X_2.

(a) Prove that Y also has precisely two irreducible components Y_1 and Y_2.

(b) Prove that the algebraic sets $X_1 \cap X_2$ and $Y_1 \cap Y_2$ are isomorphic.

(c) Are the algebraic sets $\mathcal{V}(x^2 - y^2)$ and $\mathcal{V}(x^2 - 1)$ in $\mathbb{A}^2$ isomorphic?

Solution. Let $\mathbb{C}[X]$ and $\mathbb{C}[Y]$ be the coordinate algebras of X and Y. By the assumption these are isomorphic $\mathbb{C}$-algebras. Let us choose an isomorphism $\phi : \mathbb{C}[X] \xrightarrow{\sim} \mathbb{C}[Y]$.

(a) We are given that X has two irreducible components X_1 and X_2. By one of the definitions of irreducible components, this means that the (unique!) primary decomposition of the ideal $(0) \lhd \mathbb{C}[X]$ is $(0) = P_1 \cap P_2$ where P_1 and P_2 are the incomparable prime ideals $\mathcal{I}(X_1)$ and $\mathcal{I}(X_2)$, respectively. Applying the isomorphism ϕ, we deduce that the primary decomposition of $(0) \lhd \mathbb{C}[Y]$ is $(0) = \phi(P_1) \cap \phi(P_2)$, where again $\phi(P_1)$ and $\phi(P_2)$ are incomparable prime ideals. Hence, $Y_1 := \mathcal{V}(\phi(P_1))$ and $Y_2 := \mathcal{V}(\phi(P_2))$ are the two irreducible components of Y.

(b) The ideal $\mathcal{I}(X_1 \cap X_2) = \sqrt{P_1 + P_2}$. Similarly, $\mathcal{I}(Y_1 \cap Y_2) = \sqrt{\phi(P_1) + \phi(P_2)}$. Since ϕ is an algebra isomorphism, $\sqrt{\phi(P_1) + \phi(P_2)} = \phi(\sqrt{P_1 + P_2})$. Thus, $\phi(\mathcal{I}(X_1 \cap X_2)) = \mathcal{I}(Y_1 \cap Y_2)$. So $\phi : \mathbb{C}[X] \xrightarrow{\sim} \mathbb{C}[Y]$ induces an isomorphism

$$\bar{\phi} : \mathbb{C}[X_1 \cap X_2] = \mathbb{C}[X]/\mathcal{I}(X_1 \cap X_2) \xrightarrow{\sim} \mathbb{C}[Y_1 \cap Y_2] = \mathbb{C}[Y]/\mathcal{I}(Y_1 \cap Y_2).$$

Hence, $X_1 \cap X_2 \cong Y_1 \cap Y_2$.

(c) Note that $X = \mathcal{V}(x^2 - y^2)$ has two irreducible components $X_1 = \mathcal{V}(x - y)$ and $X_2 = \mathcal{V}(x + y)$, and $Y := \mathcal{V}(x^2 - 1)$ has two irreducible components $Y_1 = \mathcal{V}(x - 1)$ and $Y_2 = \mathcal{V}(x + 1)$. Since $X_1 \cap X_2 = \{0\}$ and $Y_1 \cap Y_2 = \varnothing$, the algebraic sets $X_1 \cap X_2$ and $Y_1 \cap Y_2$ are not isomorphic. Hence, by (b), X and Y cannot be isomorphic either.

Comment. The solution to (a) and (b) above is expressed in terms of coordinate algebras. It could also be explained in more geometric terms as follows. Let $f : X \to Y$ be an isomorphism of algebraic sets. Thus, it is a bijective morphism of affine varieties whose inverse is also a morphism of affine varieties.

In particular, since morphisms of affine varieties are continuous maps for the Zariski topology, f is a homeomorphism.

(a) By another version of the definition, the irreducible components of X are its maximal irreducible subsets in the Zariski topology. We are given that there are exactly two, namely, X_1 and X_2. Applying the homeomorphism f, we deduce that $Y_1 := f(X_1)$ and $Y_2 := F(X_2)$ are the maximal irreducible subsets of Y. Hence, Y has exactly two irreducible components.

(b) We have that $Y_1 \cap Y_2 = f(X_1) \cap f(X_2) = f(X_1 \cap X_2)$. Hence, $f : X \to Y$ restricts to a bijective morphism of affine varieties $\bar{f} : X_1 \cap X_2 \to Y_1 \cap Y_2$. It is an isomorphism because its two-sided inverse is the restriction of $f^{-1} : Y \to X$, which is also a morphism of affine varieties.